

Lucas Chiang

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EDUCATION

Purdue University

Bachelor of Engineering in Electrical Engineering; Minors in Computer Engineering and Management

West Lafayette, Indiana

Class of 2030

SKILLS

Hardware: PCB Design and Manufacturing, Digital Logic Design, Circuit Design, Integrated Circuits, Materials Characterization

Programming Python, C, SystemVerilog, MATLAB, Java

Tools: Linux, Fusion 360, STK, AstroImageJ

RESEARCH EXPERIENCE

mmhLabs (DREAM Lab)

Research Intern

August 2026 – Current

West Lafayette, Indiana

- Analyzed 3+ Research Papers to build schematics of benchmark PCBs of different functions, including pH sensors and biomedical devices
- Conducted independent study to be able to manufacture benchmark PCBs to test the new IC platform, evaluating its potential to reduce the size of electrical components

Sustainable Wearable Electronics Researcher, UMBC

Research Intern

July 2025 – August 2026

Baltimore, Maryland

- Designed a PCB/IC platform on a breadboard to be able to generate an electrical pulse every few minutes using thermoelectric generators (TEGs), enough to power a wearable electrocardiograph for a short time to receive a reading
- Tested 10 different trials with 4 different sets of TEGs, along with 3+ controlled variables in the testing environment to determine best array of TEGs for efficient voltage generation

Thin Films Researcher, Towson University

Research Intern

July 2025 – August 2025

Towson, Maryland

- Fabricated 10 CMO thin-film metal oxide samples using Pulsed Laser Deposition and measured electrical resistance using the four-point probe method
- Attained ISO 7 clean room experience for fluorination of 5+ thin films
- Collaborated with graduate researchers to characterize structural properties and correlate them with electrical transport behavior using X-Ray diffraction

NASA Data Science Analysis and Scholarly Research Paper Writing Internship

Research Intern

Summer 2024

Fairfax, Virginia

- Examined ground-based light-curve data from the George Mason Telescope using a Python-based graphical system AstroImageJ to validate a TESS Object of Interest as a candidate exoplanet
- Designed a Python Jupyter Notebook program to format 100+ images into light curve data and analyze the data as the preliminary review of the exoplanet TOI 5868.01

TECHNICAL EXPERIENCE

Purdue System-on-Chip Extension Technologies

STARS Chip Design Program

August 2026 – Current

West Lafayette, Indiana

- Built technical foundations in digital logic, circuit design, transistors, and SystemVerilog

Purdue Orbital

Propulsion Team

August 2026 – Current

West Lafayette, Indiana

- Joined the propulsion team for Purdue Orbital

FIRST Tech Challenge (FTC) Team 18996

Design Leader and Outreach Lead

September 2020 – June 2026

Clarksville, Maryland

- Directed design-team meetings and applied risk-management methodologies to guide 30+ subsystem requirements and 20+ 3D printed and/or CNC'd prototypes each season
- Wired robot electrical systems and iterated more than 10 times on intake and transfer mechanisms; 3D-printed over half of the robot's parts using Fusion 360
- Led outreach for 3 years, securing partnerships with 10+ sponsors and teaching STEM to under served youth in Baltimore